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Urdaneta City Campus
College of Engineering and Architecture
Mechanical Engineering Department
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DESIGN OF 15MLD WASTEWATER TREAMENT PLANT IN VILLASIS, PANGASINAN

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Introduction

In the 21st century, water has emerged as one of the most precious yet most vulnerable resources on the planet. As populations surge and industrial activities expand, the volume of wastewater generated by municipalities grows exponentially, posing a severe threat to public health and the environment. Wastewater treatment—the process of removing physical, chemical, and biological contaminants from sewage and industrial runoff—is no longer a luxury but a critical infrastructure requirement. The primary objective of a modern Wastewater Treatment Plant (WWTP) is to process "influent" so that the resulting "effluent" can be safely discharged back into natural water bodies or repurposed for irrigation, thereby completing a sustainable water cycle.

For the municipality of Villasis, Pangasinan—famously known as the "Vegetable Basket of the North"—water quality is tied directly to the local economy and the health of its citizens. The town's proximity to the Agno River, one of the largest river systems in the Philippines, makes the implementation of a high-capacity treatment facility an ecological priority. Without proper treatment, nitrogen, phosphorus, and pathogenic bacteria from residential and commercial sectors can lead to eutrophication and the spread of waterborne diseases, ultimately damaging the agricultural lands that the province relies upon.

This project focuses on the design and operation of a 15 MLD (Million Liters per Day) Wastewater Treatment Plant utilizing an Activated Sludge System. From an Industrial Plant Engineering perspective, this facility is a complex assembly of mechanical systems, including high-efficiency aeration blowers, submersible pumps, automated screening mechanisms, and sedimentation tanks. The design integrates principles of fluid mechanics, thermodynamics, and biological engineering to ensure that the plant operates at peak efficiency while minimizing energy consumption.

Furthermore, this facility is designed in strict compliance with the Philippine Clean Water Act of 2004 (Republic Act No. 9275) and the standards set by the Department of Environment and Natural Resources (DENR). By implementing advanced filtration and disinfection stages, the plant ensures that the water returned to the Agno River meets or exceeds Class C water quality standards. Beyond environmental protection, this Industrial Plant Engineering study explores the optimization of plant layout, the strategic selection of heavy-duty machinery, and the management of manpower to create a facility that is both economically viable and operationally sustainable for the people of Villasis.

Through this comprehensive layout and design, the Villasis Wastewater Treatment Plant stands as a model of modern engineering—transforming a hazardous byproduct of urbanization into a clean resource that safeguards the ecological future of Pangasinan.


VILLASIS
AGNO RIVER
TO URDANETA
TO MANILA
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Strategic Plant Location

The proposed Wastewater Treatment Plant is ideally located in Villasis, Pangasinan, strategically positioned adjacent to the Agno River. This prime location provides a critical geographical advantage for the efficient discharge of treated effluent, making the entire environmental management process more effective and sustainable.

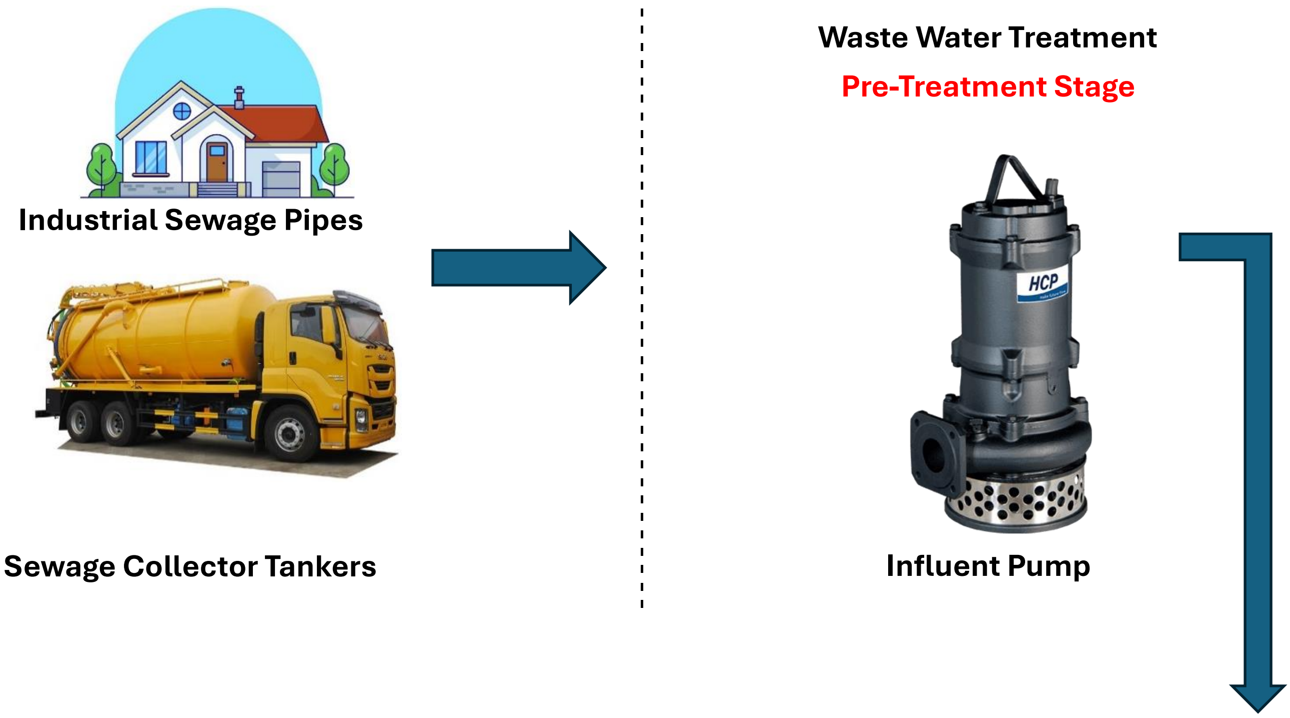
The facility's footprint is optimized within a 35,000 m² property, providing ample space for current 15 MLD (Million Liters per Day) operations while maintaining dedicated zones for future modular expansion. This large land area ensures that as the municipality grows, the plant can scale its capacity without the constraints of limited space.

With proximity to major provincial roads and the McArthur Highway, the site ensures smooth logistics for the delivery of essential treatment chemicals and the quick, cost-effective transport of treated biosolids (sludge cake). This connectivity allows the plant to serve the residents and industrial zones of Villasis in a timely and efficient manner. The combination of a well-connected location and significant land area makes this site an ideal base for safeguarding the ecological health of the region and ensuring the continued success of the municipality's infrastructure.

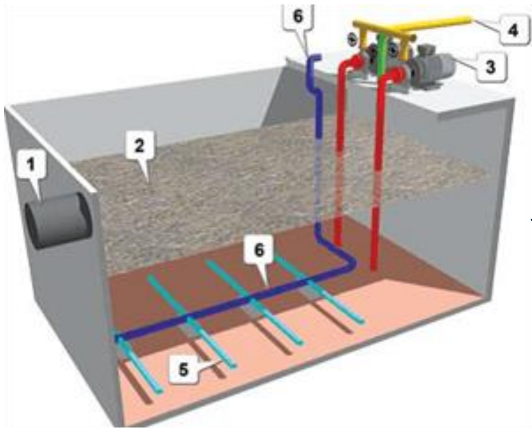
Process Flow Diagram

The Main Process

The treatment process follows a continuous flow designed to remove physical, chemical, and biological pollutants:

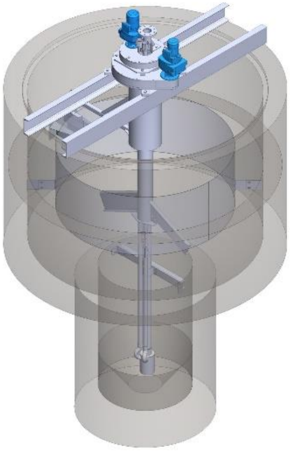


Mechanical Screen Chamber



Balancing Tanks

Primary Stage



Grit Chamber

Primary Clarifier

Secondary Stage

Liquid

Sediments

Tertiary Stage

Aeration Basins

Return Activated Sludge

Secondary Clarifiers

Clarified Water

Activated Sludge

TERTIARY STAGE

Tertiary Stage (Liquid)

Clarified Water

UV Light Disinfection Channel

Chlorination tank

River System

Tertiary Stage (Solid)

Activated Sludge

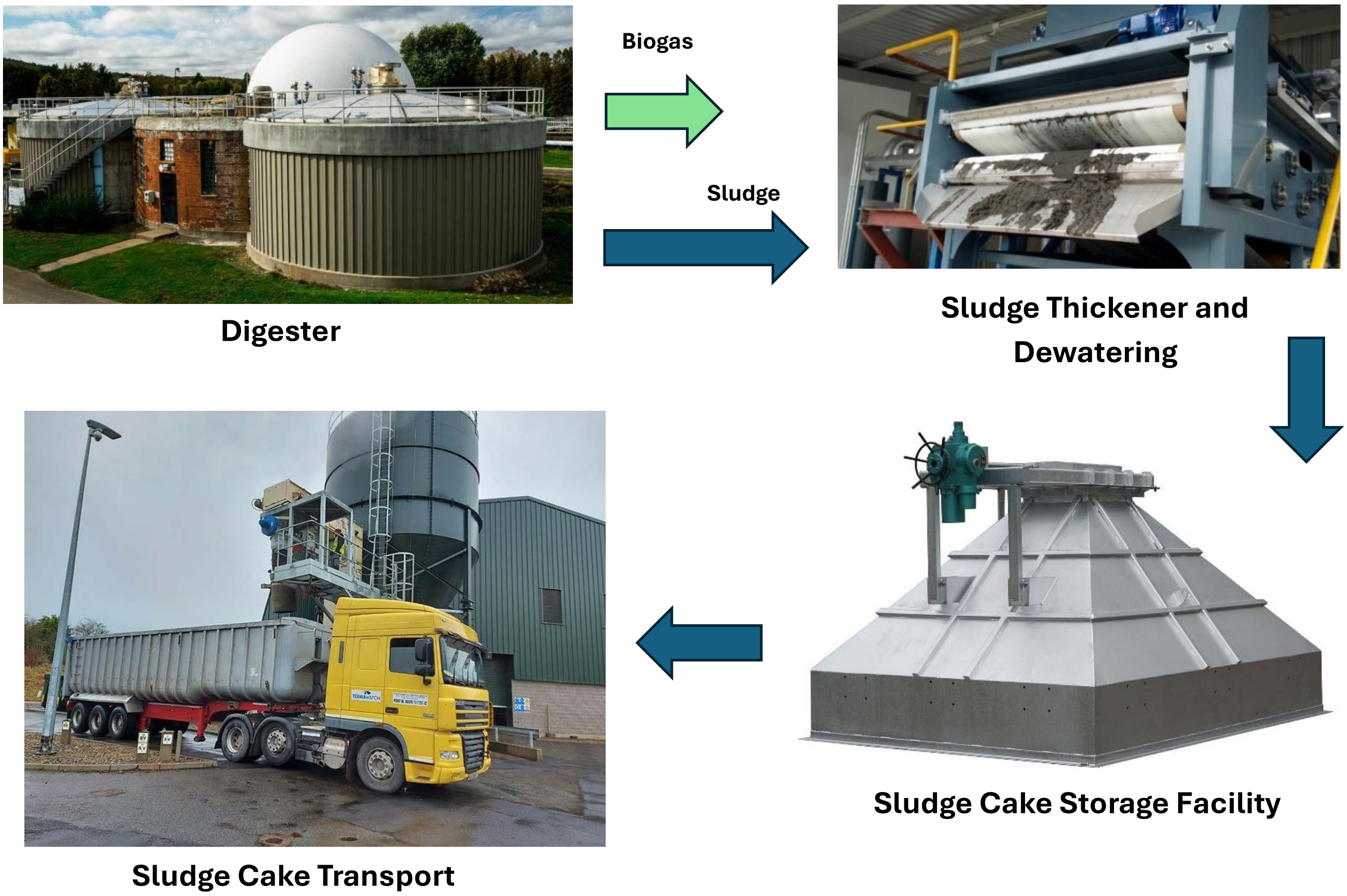
Return Activated Sludge

Aeration Basins

Waste Activated Sludge

Sludge

Digester



Preliminary Stage

Effluent Pump: Raw sewage from the Villasis municipal collection network arrives at the plant's lowest elevation. Heavy-duty, non-clog submersible centrifugal pumps lift the influent to the headworks, providing the necessary potential energy for gravity flow through the rest of the plant.

Banker Tanks: These tanks act as a buffer to manage hydraulic surges. By equalizing the flow and strength of the wastewater, they ensure a consistent loading rate into the biological stages, preventing system shock during peak morning or evening hours.

Screen Bar Chambers: Automated mechanical bar screens intercept large objects such as plastics, rags, and wood. A rake mechanism periodically clears the screens, depositing debris into a conveyor for proper solid waste disposal.

Grit Chamber: This unit reduces the velocity of the flow to allow heavy inorganic solids—such as sand, silt, and gravel—to settle at the bottom. This prevents abrasive wear on downstream pumps and prevents the accumulation of inert material in the aeration basins.

Primary Stage

Primary Clarifier: The wastewater enters large circular sedimentation tanks where the flow velocity is significantly reduced. Suspended organic solids settle to the bottom as "primary sludge," while fats, oils, and grease (scum) float to the surface. Mechanical scrapers continuously collect the sludge for treatment, while rotating skimmers remove the surface scum.

Secondary Stage

Aeration Basins: In these high-capacity basins, air is vigorously injected through fine-bubble diffusers located at the bottom. This provides oxygen to a dense population of aerobic bacteria (activated sludge) that consume the dissolved organic matter. This process converts pollutants into carbon dioxide, water, and additional biomass.

Secondary Clarifier: The mixture of treated water and biomass flows into final settling tanks. Here, the microorganisms settle to the bottom. A portion of this biomass is recycled back to the Aeration Basins as Return Activated Sludge (RAS) to maintain the bacterial population, while the excess is diverted as Waste Activated Sludge (WAS) for processing.

Final Stage

UV Sanitation and Chlorine: The clarified water passes through a UV radiation channel that disrupts the DNA of bacteria and viruses. As a secondary safeguard, a controlled dose of Chlorine is applied in a contact tank to maintain a residual disinfectant level.

Sludge Digester: The collected primary and waste activated sludge are sent to the digester. Through biological stabilization, the volume of the sludge is reduced, and volatile organic compounds are broken down, significantly reducing odors.

Sludge Thickener and Dewatering: The stabilized sludge is concentrated in a thickener and then processed through a mechanical Belt Filter Press. This equipment uses pressure to squeeze out remaining water, transforming liquid sludge into a semi-solid "cake."

Sludge Cake Storage: The dewatered sludge cakes are stored in a covered, ventilated facility. These nutrient-rich biosolids are periodically hauled away for use as agricultural fertilizer or disposed of in a managed landfill, completing the plant's circular waste management cycle.

Manufacturing Process Flow

The "manufacturing" process is the systematic transformation of raw, contaminated influent into high-quality effluent suitable for environmental discharge. This flow is categorized into sequential stages of treatment, each designed to target specific pollutants (solids, organic matter, and pathogens).

1. Preliminary Treatment Phase (Physical Separation)

The process begins with the physical removal of large objects and heavy inorganic materials to protect the mechanical integrity of the plant.

- **Influent Lifting:** Raw sewage enters the plant through the Influent Pump Station, where submersible pumps lift the water to the headworks to allow for subsequent gravity-driven flow.
- **Mechanical Screening:** The water passes through Screen Bar Chambers. Coarse screens remove large debris (rags, plastics), followed by fine screens that capture smaller particulates.
- **Degritter Process:** In the Grit Chamber, the flow velocity is carefully controlled. Heavy inorganic materials like sand and gravel settle to the bottom and are removed by a grit screw or pump, preventing abrasive wear on downstream equipment.

2. Primary Treatment Phase (Sedimentation)

This stage focuses on removing settleable organic solids through the principle of gravity.

- **Primary Clarification:** The screened water enters large, circular Primary Clarifiers. Here, the velocity is drastically reduced, allowing approximately 50-70% of suspended solids to settle as "Primary Sludge." Simultaneously, surface skimmers remove floating oils and grease (scum). The clarified liquid (supernatant) then overflows into the next stage.

3. Secondary Treatment Phase (Biological Oxidation)

This is the core "manufacturing" stage where dissolved organic pollutants are biologically broken down.

- **Activated Sludge Process (Aeration):** The water enters Aeration Basins where high-efficiency blowers inject oxygen. This environment allows aerobic bacteria (biomass) to consume dissolved organic matter (BOD).
- **Biological Separation:** The mixture of water and biomass flows to Secondary Clarifiers. The biomass settles to the bottom.
    - **RAS (Return Activated Sludge):** A portion of the settled bacteria is pumped back into the Aeration Basins to maintain the biological population.

- **WAS (Waste Activated Sludge):** Excess biomass is purged from the system and sent to the solids handling line.

4. Tertiary Treatment & Disinfection Phase (Polishing)

This final stage ensures the water is chemically and biologically safe for the Agno River ecosystem**.**

- **Filtration/Polishing:** If necessary, the water passes through sand or multi-media filters to remove any remaining fine suspended solids.
- **Pathogen Neutralization:** The water enters the UV Sanitation channel or Chlorine Contact Tank. UV radiation or chemical dosing destroys the DNA of harmful microorganisms, ensuring the effluent is pathogen-free.

5. Solids Handling & Byproduct Management (Sludge Line)

The pollutants removed in the previous stages are processed into a stable, manageable form.

- **Thickening & Digestion:** Combined sludge from primary and secondary stages is sent to the Sludge Thickener to reduce water volume, then to a Digester for biological stabilization and odor reduction.
- **Dewatering:** The stabilized sludge is processed through a Belt Filter Press or Centrifuge, resulting in a semi-solid Sludge Cake.
- **Final Disposal:** The nutrient-rich sludge cakes are moved to Sludge Cake Storage, where they are prepared for transport as agricultural soil conditioner or for disposal in a sanitary landfill.

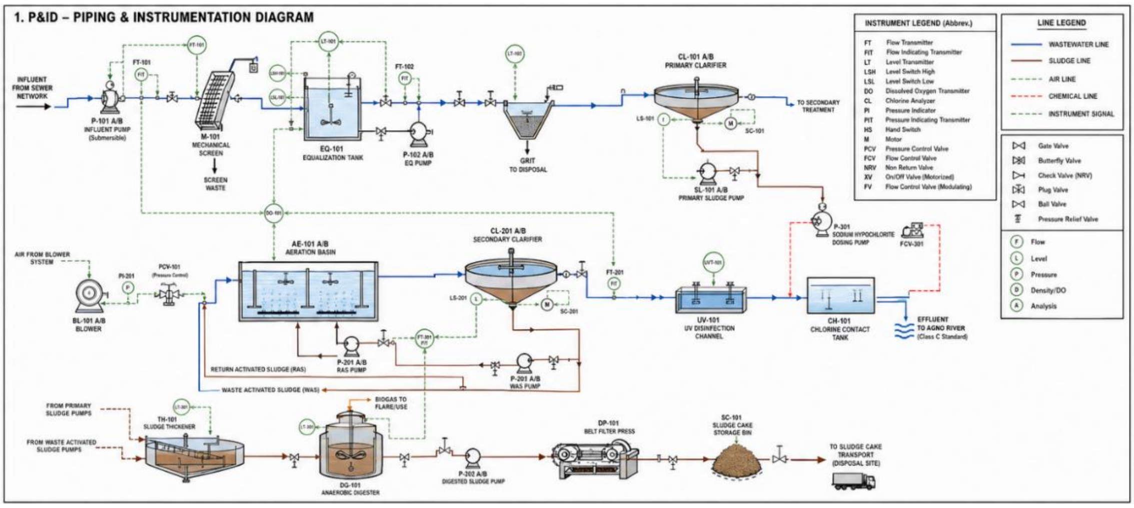
Piping and Instrumentation Diagram

The Piping and Instrumentation Diagram (P&ID) of the 15 MLD wastewater treatment plant illustrates the complete integration of hydraulic flow, mechanical equipment, and control systems in an activated sludge process. Raw influent is first lifted by submersible pumps and directed through mechanical bar screens and an equalization tank, where flow and pollutant loading are stabilized before entering the grit chamber for removal of heavy inorganic materials.

The wastewater then proceeds to the primary clarifier, where settleable solids are separated as sludge and conveyed to the sludge handling system, while clarified liquid moves into the aeration basin. In this biological treatment stage, air supplied by centrifugal blowers is regulated through pressure control valves and fine bubble diffusers, with dissolved oxygen levels continuously monitored and controlled to sustain microbial activity. The mixed liquor then flows into the secondary clarifier, where biomass is settled and either returned to the aeration basin as return activated sludge (RAS) or removed as waste activated sludge (WAS) for further treatment.

The clarified effluent undergoes tertiary treatment through UV disinfection followed by chlorination to ensure pathogen removal before discharge.

Meanwhile, collected sludge is thickened, digested for stabilization and biogas production, and dewatered using a belt filter press to produce sludge cake for disposal or reuse. Throughout the system, instrumentation such as flow transmitters, level sensors, dissolved oxygen analyzers, and chlorine analyzers are integrated into a **SCADA-controlled** network, enabling automated monitoring and control of pumps, blowers, and dosing systems, thereby ensuring efficient, safe, and compliant plant operation.



Plant Layout

The proposed plant layout design for the 15 MLD Wastewater Treatment Plant in Villasis, Pangasinan, aimed at ensuring a continuous hydraulic flow, optimal equipment placement, and strict compliance with environmental and safety standards. The layout encompasses critical zones such as preliminary screening, biological aeration, secondary clarification, and sludge processing, strategically positioned to enhance treatment efficiency while maintaining rigorous hygiene and odor control protocols. This design aligns with DENR Administrative Order (DAO) guidelines and supports the municipality's goal of protecting the Agno River with maximum operational cost-efficiency.

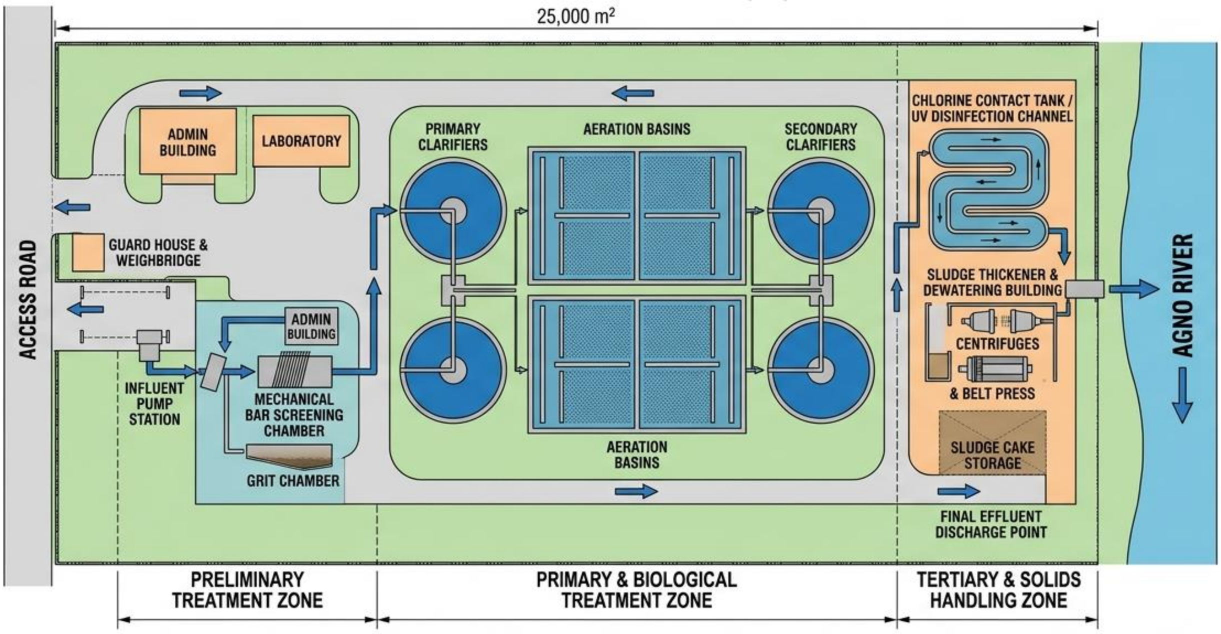
Process	Approx. Area Allocation	Remarks
1. Influent Pump Station	1,500 m^2	Inlet works, lifting raw sewage
2. Mechanical Screening	1,000 m^2	Automated bar screens, debris removal
3. Grit Chamber	800 m^2	Removal of sand, silt, and stones
4. Primary Clarifiers	2,500 m^2	Initial sedimentation, scum removal
5. Aeration Basins	5,000 m^2	Core biological zone, fine bubble diffusers
6. Secondary Clarifiers	2,500 m^2	Biomass settling, RAS/WAS separation
7. Disinfection Channel	1,500 m^2	UV radiation and Chlorine contact
8. Sludge Thickening	2,000 m^2	Gravity thickeners for waste sludge

Process	Approx. Area Allocation	Remarks
9. Dewatering Building	3,000 m²	Belt filter press, centrifuge area
10. Sludge Cake Storage	3,200 m²	Covered area for dried biosolids
Support/Admin/QA Lab	1,500 m²	Offices, monitoring, water testing
Infrastructure/Expansion	**35,000 m² Total**	**Includes roads, buffer, and Phase 2 plots**

The design of the Villasis Wastewater Treatment facility's floor plan is based on a logical, hydraulic-driven flow that maximizes treatment efficacy and minimizes energy consumption—critical factors in municipal infrastructure. The layout starts with the Influent Pump Station placed near the facility's lowest elevation point, ensuring the seamless intake of raw municipal sewage. Adjacent to it is the Preliminary Treatment Zone (Screening and Grit Removal), enabling quick and organized removal of physical debris and abrasive materials using automated rakes and grit classifiers.

Next, the flow moves into the Primary Clarifiers and then the Aeration Basins, which are strategically positioned at the core of the operation. This central positioning allows for the most intensive mechanical processes—such as high-capacity oxygen injection—to be contained in a single managed block, promoting a continuous biological oxidation flow into the Secondary Clarification stage without hydraulic backtracking.

Downstream, the Disinfection Channel and Final Effluent Point are aligned consecutively to ensure the treated water is pathogen-free before reaching the facility's outfall at the Agno River. The Sludge Handling Complex (Dewatering and Storage) is intentionally segmented at the far end of the property. This separation reduces odor interference with the administrative area and provides a dedicated heavy-vehicle access route for the transport of stabilized sludge cakes. The overall layout uses a linear progression to guide the directional flow of wastewater, ensuring environmental compliance (especially between influent and high-quality effluent) and supporting the long-term scalability of the Villasis water management network.



List of Manpower

The successful execution and continuous operation of the 15` Wastewater Treatment Plant in Villasis, Pangasinan, rely heavily on the collaborative efforts of a skilled multidisciplinary team. This section outlines the manpower assigned to various phases of the plant's operation, detailing their roles and responsibilities to ensure compliance with DENR standards and optimal mechanical performance. Each team member is selected based on technical expertise in environmental engineering, mechanical systems, and chemical processing.

Top Management

- Board of Directors
- Chief Executive Officer (CEO)
- Chief Operating Officer (COO)
- Chief Technical Officer (CTO)

1. Influent Pump Station & 2. Mechanical Screening

Department: Intake Operations & Preliminary Treatment

- **Operations & Maintenance Director**
    - Inlet Works Supervisor
        - Pump Station Technicians
        - Mechanical Screen Operators
        - Heavy Equipment Mechanics
        - Debris Handling Staff
    - Lift Station Coordinator

3. Grit Chamber & 4. Primary Clarifiers

Department: Physical Treatment & Sedimentation

- **Plant Manager**
    - Primary Treatment Supervisor
        - Grit Classifier Operators
        - Sedimentation Technicians
        - Scum & Grease Collectors
    - Maintenance & Lubrication Specialist

5. Aeration Basins & 6. Secondary Clarifiers

Department: Biological Process & Mechanical Systems

- **Process Engineering Director**
    - Biological Systems Supervisor
        - Aeration Blower Operators
        - Dissolved Oxygen (DO) Monitoring Technicians
        - Secondary Clarifier Operators
    - SCADA & Automation Engineer
    - RAS/WAS Pump Technicians

7. Disinfection Channel (UV & Chlorine)

Department: Tertiary Treatment & Water Quality

- **Environmental Compliance Director**
    - Disinfection Supervisor
        - UV System Maintenance Technicians
        - Chemical Dosing Operators
        - Final Effluent Sampling Officers

8. Sludge Thickening & 9. Dewatering Building

Department: Solids Handling & Residuals Management

- **Manufacturing & Solids Manager**
    - Sludge Processing Supervisor
        - Centrifuge / Belt Press Operators
        - Sludge Thickener Technicians
        - Polymer Dosing Specialists
    - Process Efficiency Engineer

10. Sludge Cake Storage & Disposal

Department: Logistics & Biosolids Distribution

- **Logistics & Disposal Manager**
    - Storage & Loading Supervisor
        - Heavy Equipment / Forklift Operators
        - Sludge Hauling Coordinators
        - Inventory & Dispatch Clerks

Quality Assurance & Quality Control (QA/QC)

- **QA/QC Director**
    - Laboratory Manager
        - Environmental Chemists
        - Microbiologists (Biomass Analysis)
        - Water Quality Inspectors
    - Compliance & Validation Officer

Regulatory Affairs & Environmental Compliance

- **Regulatory Affairs Director**

- DENR Compliance Manager
  - Pollution Control Officer (PCO)
  - Permit & Documentation Specialist
  - Environmental Audit Coordinator

Sales & Public Relations (Utility Management)

- **Marketing & External Affairs Director**
  - Public Relations Manager
    - Community Liaison Officers
    - Industrial Account Specialists
  - Market Analyst (Water Reuse Research)

Human Resources & Administration

- **HR Director**
  - HR Manager
    - Safety & Health Officer (Safety Engineer)
    - Recruitment & Training Officer
    - Payroll & Timekeeping Specialist

Finance & Accounting

- **Finance Director**
  - Chief Accountant
    - Budget & Cost Control Analyst (Chemical/Power Expenses)
    - Accounts Payable/Receivable Officer
    - Procurement & Sourcing Officer

The Capacity of the Plant

The Villasis Wastewater Treatment Plant is engineered with a robust hydraulic capacity of 15,000 cubic meters per day (15 MLD), ensuring consistent processing of municipal influent and the ability to meet the high-volume demands of a growing population and industrial sector. This capacity is designed to maintain optimal hydraulic retention times, ensuring that all discharged effluent strictly adheres to environmental safety standards.

Factor	Value
Plant Site Area	25,000 m^2 (2.5 Hectares)
Daily Capacity	15,000 m^3/day (15 MLD)
Annual Capacity	~5.47 Million m^3/year
Outputs (Products)	Treated Effluent (Class C), Stabilized Sludge Cake
Operational Basis	24/7 Continuous Flow, 4 Aeration Basins, 2 Clarifier Trains

List of Equipment and its Specifications

The primary mechanical and electrical equipment required for the operation of the 15 MLD Wastewater Treatment Plant. The selection of equipment is based on durability, energy efficiency, and compliance with industrial standards for continuous 24/7 operation in a corrosive environment.

Preliminary Treatment Equipment

Equipment Name	Technical Specifications	Qty	Purpose
Submersible Influent Pumps	Capacity: 350 m^3/h each; Head: 20m; Power: 30 kW; Material: Cast Iron/SS316 Impeller	3	Lifts raw sewage to the plant headworks (2 duty, 1 standby).

Equipment Name	Technical Specifications	Qty	Purpose
Mechanical Bar Screen	Type: Multi-rake; Bar Spacing: 6mm; Channel Width: 1200mm; Material: SS304; Motor: 1.5 kW	2	Automatically removes large debris and solids from the influent.
Vortex Grit Classifier	Capacity: 800 m^3/h; Material: SS304; Drive: 2.2 kW; Efficiency: 95% removal of 0.2mm grit	1	Separates and removes sand, silt, and heavy inorganic matter.

Submersible Influent Pumps

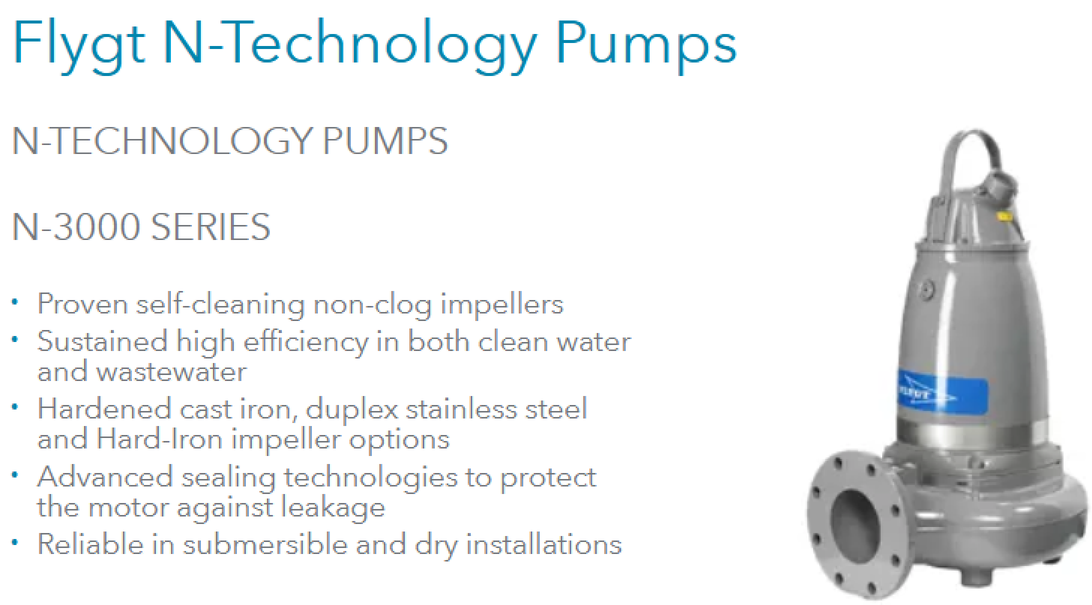


Figure: Xylem N3000 Series Pump

This type of pump handles capacities up to 1,760 l/s (26,600 gpm) with motors from 1.3 to 680 kW (1.5 to 870 hp) and pumping heads up to 100 m (350 ft). Choose a hardened cast iron impeller for typical wastewater applications or the chopper version for your toughest applications with long fibers and heavy solids. Hard-Iron impellers are highly recommended for applications with sand, grit and other abrasives, as well as wastewater mixed with sea water. For special applications such as highly corrosive wastewater environments we offer impellers in duplex stainless steel.

Mechanical Bar Screen

Figure: HUBER Muti-Rake Bar Screen

The system is designed to deliver very low headloss while maintaining high separation efficiency, ensuring optimal hydraulic performance. Its defined meshing of the cleaning rakes with the bar rack enhances operating reliability, while the design allows installation without the need for a bottom step. The unit features a compact structure with minimal installation height above ground level, along with a completely odour-encased screen equipped with easily removable covers for maintenance. It can be conveniently retrofitted into existing channels without requiring channel recesses, and the self-supporting folded stainless steel profile enables easy lifting from the channel. The screen operation is not hindered by gravel or grit, and it includes a simple, easily accessible chain tensioning unit for maintenance efficiency. All components in contact with the medium—except the chain, pinion, drive, and bearing—are constructed from immersion pickled stainless steel, with optional stainless steel chains and chain wheels available. The system also provides high screenings discharge capacity through adjustable cleaning elements, with rake and comb plates that can be replaced independently. Additionally, it features an automatic scraper mechanism that operates without the need for service water, further simplifying operation and maintenance.

Vortex Grit Removal

Figure: PISTA Invorsor Grit Removal System

The **INVORSOR**® delivers lower capital and operational costs, larger capacity in individual units, greater design flexibility for inlet-outlet design options, and a higher surface area-to-volume ratio to generate consistent fine grit capture during low flow, daily flow and peak flow conditions — up to 50 MGD in single units [190,000 m3/d].

Primary and Secondary Treatment Equipment

Equipment Name	**Technical Specifications**	**Qty**	**Purpose**
Primary Clarifier Scraper	Diameter: 20m; Drive: Peripheral; Material: SS304/Hot-dip Galvanized; Speed: 0.03 RPM	2	Collects settled primary sludge and removes surface scum.
Centrifugal Aeration Blowers	Type: Multi-stage Centrifugal; Flow: 2,500 m^3/h; Pressure: 0.6 bar; Motor: 75 kW with VFD	3	Provides high-volume oxygen to the aeration basins for biological oxidation.
Fine Bubble Diffusers	Type: Disc Diffuser; Material: EPDM Membrane; Diameter: 250mm; Air Flow: 2–6 m^3/h per unit	1,200	Evenly distributes oxygen into the wastewater to support aerobic bacteria.
Secondary Clarifier Drive	Diameter: 25m; Type: Half-bridge scraper; Material: SS304; Power: 3.0 kW	2	Separates treated effluent from the activated sludge mass.

Clarifier Scraper Mechanisms

Figure x: Clarifier Scraper Mechanisms

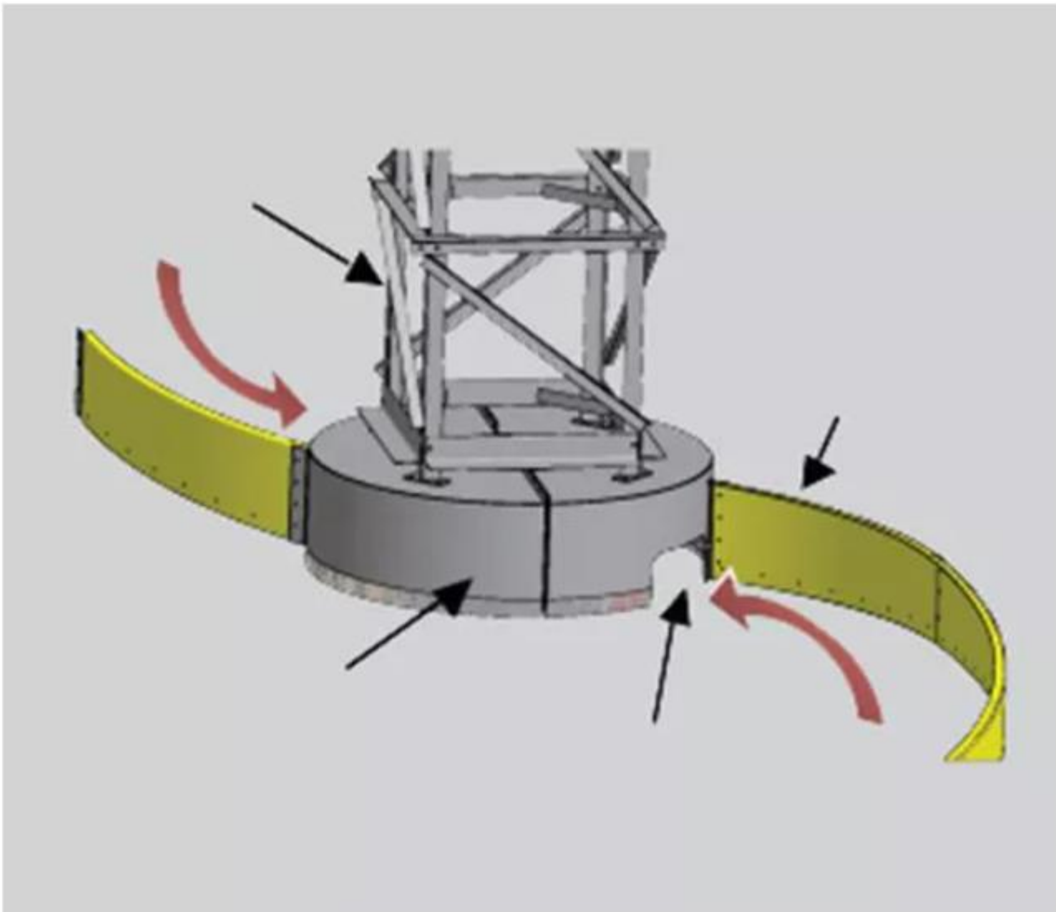


Figure: HUBER Clarifier Scraper

The Spiral Blade Scraper Clarifier is designed for ease of installation and operation, making it a practical choice for wastewater treatment applications. It is available in a wide range of sizes, from 6 to 76 meters (20 to 250 feet) in diameter, allowing flexibility for various plant capacities. The drive system features a robust design, including a cast-iron base, a time-tested strip liner bearing, a solid main gear, oil bath lubrication, and a deep oil reservoir, all contributing to long service life and reliable performance. It offers a drive torque capacity ranging from 12,202 to 1,355,818 joules (9,000 to 1,000,000 foot-pounds), designed to meet AGMA standards for 20 years of continuous operation.

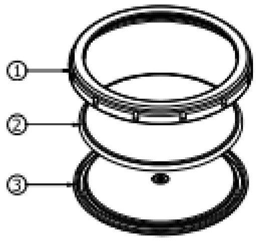
The clarifier can be fabricated using a variety of materials such as A36 steel, hot-dip galvanized steel, painted steel, and 304 or 316 stainless steel, with additional special material options available depending on operational requirements. It also provides multiple scum skimming design options, including a conventional scum box, a full-trough skimmer, and a rotating pipe skimmer, ensuring effective removal of floating solids under different process conditions.

Fine Bubble Disc Diffusers

Figure : SSI Aeration Fine Bubble Disc Diffusers

Diffuser Parts	Material
1. Retainer nut	PP
2. Membrane	EPDM / PTFE / PEEK / Silicone
3. Diffuser Base	PP

Max. Temperature resistance - 212º F / 100º C



Avg Operating Condition per Diffuser	Peak Air Flow per Diffuser	Orifice Size	Slit Qty	Active Surface Area per Diffuser	Weight per Diffuser AFD270
0.45 - 2.5 SCFM 0.8 - 4.5 Sm^3/hr	4.0 SCFM 7.0 Sm^3/hr	1/4" 6mm	+/- 6600	0.41 ft^2 0.038 m^2	1.5 lbs 0.68 kg

Figure x: Equipment Specification – 9" Fine Bubble Diffusers

Tertiary and Sludge Handling Equipments

Equipment Name	Technical Specifications	Qty	Purpose
UV Disinfection System	Type: Low-pressure high-output lamps; Capacity: 700 m^3/h; Material: SS316L Chamber; UV Dosage: 30 mJ/cm^2	1	Neutralizes pathogens and bacteria in the final effluent.
RAS/WAS Pumps	Type: Horizontal Split Case; Capacity: 150 m^3/h; Head: 12m; Power: 11 kW; Material: SS316	4	Returns activated sludge to aeration or wastes it to the sludge line.
Belt Filter Press	Belt Width: 2000mm; Capacity: 10–15 m^3/h (sludge); Material: SS304; Power: 5.5 kW	1	Dewaters stabilized sludge into semi-solid sludge cakes.
Polymer Dosing System	Tank Volume: 1000L; Pump Type: Diaphragm; Capacity: 0–50 L/h; Material: PVC/PE	2	Injects coagulants to facilitate sludge flocculation and dewatering.

UV Disinfection System

Figure x: UV Disinfection System – Trojan Technologies

The way in which water enters the UV system is controlled in such a way that stringent tolerances on concrete channel walls are not required, making chlorine contact tank and UV channel retrofits simpler. Retrofits can accommodate existing water level profile and head loss. Water level is managed using light locks in combination with the downstream water level controller. Light locks eliminate the need for complex channel designs, such as stepped channels, and help provide the highest treatment level over a wide range of flow rates and water levels.

Polymer Dosing System

Figure: Hollyep Polymer Dosing System

By using the model HLJY1000 Model. The specification of the dosing system are the following; 1000 L/H Capacity, with dimensions of 1000x1625x1750 (mm), Powder Conveyor Power of 0.37KW, Paddle Dia of 200(mewmm), Mixing Spindle Speed of 120 r/min, with a Rating power of 0.2*2 with inlet and outlet dia of 25mm.

Belt Filter Press

Figure: Komline - Kompress Belt Filter Press

A Belt Filter Press is a sludge dewatering device that applies mechanical pressure to a chemically conditioned slurry, which is sandwiched between two (2) tensioned belts, by passing those belts through a serpentine of decreasing diameter rolls. The machine can actually be divided into three (3) zones: the gravity zone, where free draining water is drained by gravity through a porous belt; the wedge zone, where the solids are prepared for pressure application; and the pressure zone, where medium, then high pressure is applied to the conditioned solids.

Supporting Systems

1. **SCADA Control System:**

   Specification: PLC-based (Programmable Logic Controller) with HMI (Human Machine Interface) touchscreen.

   Purpose: Centralized monitoring of DO levels, flow rates, pump status, and automated alarming for the entire Villasis facility.

2. **Emergency Standby Generator:**

Specification: 500 kVA, Diesel-powered, Sound-attenuated enclosure, Automatic Transfer Switch (ATS).

Purpose: Ensures continuous plant operation and prevents untreated discharge during municipal power outages.

3. **Laboratory Equipment:**

   Specification: BOD/COD Incubators, Spectrophotometer, pH Meters, TSS Filter Apparatus.

   Purpose: Daily testing of water quality parameters to ensure compliance with DENR effluent standards.

Design Calculations

All of Equipments

1. Influent Pump Power Requirement

   Design Flow (*Q*): 15,000 m³ /Day = 0.1736 m³/s

   Total Dynamic Head (*H*): 20 meters

   Pump Efficiency: 75%

   Calculating Hydraulic Power:

$$P_h = \rho \cdot g \cdot Q \cdot H$$

$$P_h = 1000\ \mathrm{kg/m^3} \cdot 9.81\ \mathrm{m/s^2} \cdot 0.1736\ \mathrm{m^3/s} \cdot 20\ \mathrm{m} = 34,060\ \mathrm{Watts} \approx 34.06\ \mathrm{kW}$$

   Calculating Brake Horsepower

$$BHP = \frac{P_h}{\eta} = \frac{34.06\ \mathrm{kW}}{0.75} = \mathbf{45.41}\ \mathrm{kW}$$

   With an BHP of approximately **61HP, a 75 HP** motor is recommended to provide the factor of safety for fluid density variations.

2. **Aeration Basin Volumetric Sizing**

   Hydraulic Retention Time: 8 Hours

   Volume (*V*) = $Q_{avg} \cdot HRT = 625\ \mathrm{m^3/hr} \cdot 8\ \mathrm{hr} = \mathbf{5,000}\ \mathrm{m^3}$

   Dimensions: Two Basins at 2500 m³ each

   (25m x 20m x 5m depth)

Equipment Foundation

1. **Mass Ratio Rule**

   The concrete foundation mass (Mf) should be at least 5 times the mass of the rotating machine to neutralize vibrations.

2. **Soil Bearing Pressure**

For a concrete block 2m x 3m x 1m:

Centrifugal Blower Weight = 1200kg

Concrete Block: 6m³ x 2400kg/m³= 14,400 kg

Total Weight (14,400 + 1200) x 9.81 = 1533,035 N

Pressure Calculation:

$$\sigma = \frac{W_{total}}{Area} = \frac{153\ \text{kN}}{6\ \text{m}^2} = \mathbf{25.5}\ \text{kPa}$$

Therefore:

It is safe as it remains below the typical **150KPA** allowable bearing capacity of local soils.

Piping and Accessories

1. **Main Influent Header Diameter**

Sized for a peak flow of 0.26 m^3 and a design velocity of 1.2m/s

$$D = \sqrt{\frac{4Q}{\pi v}} = \sqrt{\frac{4 \cdot 0.26}{\pi \cdot 1.2}} \approx 0.525\ \text{m} \rightarrow \mathbf{600}\ \text{mm (Nominal Pipe Size)}$$

2. **Head Loss (Hazen Williams)**

To Calculate the pressure drop across the 100m pipe (C=140)

$$h_f = 10.67 \cdot L \cdot \left(\frac{Q}{C}\right)^{1.85} \cdot D^{-4.87}$$

$$h_f = 10.67 \cdot 100 \cdot \left(\frac{0.26}{140}\right)^{1.85} \cdot 0.60^{-4.87} \approx \mathbf{0.28}\ \text{meters per 100m}$$

Material Handling Units

1. **Clarifier Scraper Mechanisms (Primary and Secondary)**

For a 25m diameter unit

Peripheral Speed: 3m/min (in order to prevent re-suspension)

Rotational Speed (N): $N = \frac{v_p}{\pi D} = \frac{3}{\pi \cdot 25} = \mathbf{0.038}\ \mathrm{RPM}$

Running Torque (T) $T = c \cdot D^2 = 15 \cdot 25^2 = 9,375$ kg-m $\approx \mathbf{92,000}$ N-m

Drive Power (P): At 60% gear efficiency:

$$P = \frac{2\pi \cdot N \cdot T}{60 \cdot 0.60} \approx \mathbf{0.61}\ \mathrm{kW}$$

2. **Sludge Cake Logistics:**

Calculation of transport requirements for dewatered sludge

Daily Mass: 22,500 kg/ day

Daily Volume: 22,500/1,100 kg/m³ = 20.5 m³/day

Handling Unit: Forklift with 1m³ Self- Dumping Skip Bin

Trips per 8-Hour Shift: 21 Trips = 3 Trips per hour

Definition of Terms

Activated Sludge Process: A biological treatment method where a mass of aerobic bacteria (biomass) is used to consume and break down organic pollutants in the wastewater.

Aeration: The process of injecting air (oxygen) into basins to support the growth and activity of aerobic microorganisms that treat the water.

BOD (Biochemical Oxygen Demand): A measure of the amount of dissolved oxygen needed by aerobic organisms to break down organic material in a water sample.

Clarification (Sedimentation): A physical process where flow velocity is reduced to allow suspended solids to settle to the bottom by gravity and floating scum to be removed from the surface.

DAO (DENR Administrative Order): Regulatory guidelines issued by the Department of Environment and Natural Resources that set the standards for effluent quality and plant operation.

Effluent: The treated wastewater that has passed through all purification stages and is discharged into a natural water body, such as the Agno River.

Grit Removal: The preliminary stage where heavy inorganic materials like sand, silt, and gravel are removed to prevent abrasive wear on mechanical equipment.

Influent: Raw, untreated wastewater or sewage as it enters the treatment facility for processing.

MLD (Million Liters per Day): The standard unit of measurement for the plant's hydraulic capacity; this facility is designed for a 15 MLD flow.

PCO (Pollution Control Officer): A designated technical officer responsible for ensuring the plant's continuous compliance with environmental laws and DENR standards.

Preliminary Treatment: The initial physical stage of treatment focused on removing large debris and heavy solids through screening and grit removal.

RAS (Return Activated Sludge): Settled biomass from the secondary clarifier that is pumped back into the aeration basins to maintain the bacterial population.

SCADA (Supervisory Control and Data Acquisition): A centralized computer system used for the real-time monitoring and automated control of plant operations, such as pump status and dissolved oxygen levels.

Secondary Treatment: The core biological stage where dissolved organic matter is oxidized and removed using the activated sludge process.

Septage: The liquid and solid material pumped from a septic tank or "pozo negro" during cleaning operations.

Siphoning: The mechanical removal of waste from septic tanks using high-vacuum equipment (commonly referred to as "Malabanan-style" services).

Sludge Cake: The semi-solid, dewatered byproduct of the treatment process that is stable enough for transport and disposal or use as fertilizer.

Sludge Digestion: A biological stabilization process that reduces the volume and odor of collected sludge by breaking down volatile organic compounds.

Tertiary Treatment (Polishing): The final purification stage that includes advanced filtration and disinfection to ensure the water is safe for the ecosystem.

UV Disinfection: A chemical-free disinfection method that uses ultraviolet radiation to disrupt the DNA of pathogens, rendering them harmless.

WAS (Waste Activated Sludge): Excess biomass that is removed from the biological system and sent to the sludge handling line for processing

Company Profile

The **Tae-chnical Institute** is a forward-thinking engineering and environmental solutions firm specializing in the design, fabrication, and management of advanced wastewater treatment infrastructure. Based in the heart of the Ilocos Region, we bridge the gap between complex mechanical engineering and sustainable environmental stewardship.

Our flagship initiative, the 15 MLD Wastewater Treatment Plant in Villasis, Pangasinan that serves as a testament to our commitment to protecting the Agno River ecosystem through high-precision engineering and innovative waste-to-resource conversion.

Vision

To be the Philippines' most resilient and innovative leader in liquid waste management, transforming the "unthinkable" into sustainable resources for a greener tomorrow.

Mission

To provide municipal and industrial sectors with high-efficiency treatment solutions that exceed regulatory standards, utilizing state-of-the-art mechanical design, SCADA-driven automation, and rigorous quality control.

Motto

We don’t just flush it, we study it.

Business Divisions

1. Industrial Plant Engineering (The "Institute" Division)

Our high-level engineering wing focuses on the design, calculation, and fabrication of large-scale treatment infrastructure.

- **WWTP Design:** Specialists in biological aeration, clarifier mechanics, and SCADA-integrated systems.
- **Design-Build Services:** From the 15 MLD Villasis project to municipal grit removal systems.

2. Vacuum Truck Services & Septic Management (The "Sipsip" Division)

The "boots on the ground" arm of our company provides immediate sanitation relief to residential, commercial, and industrial clients.

- **Septic Tank Siphoning:** High-vacuum removal of sludge and septage using our fleet of specialized tank trucks.
- **Manual Cleaning & Excavation:** Professional clearing of "pozo negro" (septic tanks) and removal of hardened waste.
- **Pipe & Sewer Declogging:** Utilizing high-pressure water jetting to clear municipal and residential blockages.
- **Emergency Response: 24/7 "Sipsip"** services for overflowing tanks to prevent environmental contamination.

Core Competencies

At The Tae-chnical Institute, we don't just move water; we engineer solutions. Our expertise spans:

- **Mechanical Systems Design:** Specializing in pump hydraulics, gear reductions, and precision aeration systems.
- **Wastewater Infrastructure:** Full-scale design and layout of Preliminary, Secondary, and Tertiary treatment phases.
- **Solids Management:** Innovative sludge dewatering and byproduct logistics, focusing on sustainable biosolids disposal.
- **Sustainable Engineering**: Research and development into converting waste materials (such as HDPE) into functional industrial products.
- **Project Leadership:** Strategic management of provincial engineering conferences and technical affairs.

Service Portfolio

Service Type	Target Client	Key Technical Feature
Residential Siphoning	**Households**	**Rapid-response vacuum trucks; deep-tank cleaning.**
Industrial Tankering	**Factories/Malls**	**Large-volume haulage (up to 10,000L per trip).**
Infrastructure Design	**Municipalities**	**Hydraulic design of treatment plants and lift stations.**
Grease Trap Maintenance	**Restaurants/Food Hubs**	**Removal of FOG (Fats, Oils, and Grease) to prevent sewer clogs.**

Contact & Operations
The Tae-chnical Institute

Owner: John Carlo M. Rebalbusa

Plant Location: Puelay, Villasis, Pangasinan

Main Office: Poblacion Zone I, Villasis, Pangasinan

***Service Area:* Villasis, Urdaneta, and throughout the Pangasinan Region.**

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